

APEX UNIVERSITY • DEPARTMENT OF PHILOSOPHY

PHIL 385: Ethics of Emerging Technologies

Term Essay & Policy Analysis

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Algorithmic Utilitarianism and the Epistemic Limits of Autonomous Decision-Making

Assignment Prompt: Critically evaluate whether Act and Rule Utilitarianism provide a morally defensible framework for decision-making in autonomous algorithmic systems operating under epistemic uncertainty. Contrast utilitarian optimization with Kantian deontological side-constraints, drawing upon the fictional **Nexa Resource Allocation Protocol** case study. Propose an actionable normative architecture for algorithmic governance.

1. Introduction: The Epistemic Dilemma of Synthetic Agency

The rapid integration of automated decision-making engines into public administration, triage medicine, and financial risk assessment marks a fundamental shift in moral agency. Where normative ethics traditionally evaluated human agents possessing reflexive intentionality, modern algorithmic governance delegates life-altering choices to stochastic optimization models.¹ Because machine learning architectures excel at multi-variable objective minimization, computer scientists and policy designers frequently adopt consequentialist frameworks—specifically Act and Rule Utilitarianism—as default operational paradigms.² Under this regime, an algorithm's prospective choice is deemed morally optimal if it maximizes an aggregate welfare function, such as Quality-Adjusted Life Years (QALYs) or net economic utility.

However, this algorithmic implementation of consequentialism founders upon a deep epistemological obstacle: the impossibility of calculating future utility under systemic uncertainty. In complex societal environments, synthetic decision engines must act upon incomplete training distributions and opaque probabilistic projections. When utility functions are computed over proxy metrics, consequentialist optimization routinely produces counter-intuitive and morally impermissible outcomes.

This essay argues that while Rule Utilitarianism provides valuable initial heuristic tools for macro-level policy design, it is fundamentally inadequate as an unconstrained normative foundation for real-time autonomous systems. Through an examination of the **Nexa Resource Allocation Protocol**—a fictional healthcare triage algorithm deployed during emergency hospital overload—I demonstrate that pure utility maximization leads to systemic instrumentalization of vulnerable individuals. To prevent catastrophic moral failures, algorithmic architectures must incorporate absolute Kantian deontological side-constraints based upon the Categorical Imperative's Formula of Humanity.

2. The Mechanics and Blindspots of Algorithmic Consequentialism

Consequentialist moral theory asserts that the normative value of an action is entirely determined by its produced outcomes.³ When embedded in software systems, this framework translates into scalar utility functions ($U : S \rightarrow \mathbb{R}$), where the algorithm selects action $a^* \in \mathcal{A}$ that maximizes expected utility across system states S :

$$a^* = \arg \max_{a \in \mathcal{A}} \sum_{s \in S} P(s | a) \cdot U(s) \quad (1)$$

¹ Julian Thorne, *Moral Algorithms: Epistemology and Ethics in Synthetic Reasoning* (Apex University Press, 2024), 45–48.

² Clara Vance and Meridian Ethics Research Group, "Deontological Constraints on Algorithmic Choice," *Journal of Applied Philosophy* 41, no. 2 (2025): 114.

³ Jeremy Bentham, *An Introduction to the Principles of Morals and Legislation* (London: Payne and Foss, 1789), chap. 1.